Section 5 — Mapping to UK Observables

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Operationalises the captive-demand FTPL model of Section 4 against UK data. Four subsections: postwar repression context (1945–1980), steady-state mapping, year-by-year wedge identification, and the wedge time series with convenience-yield comparison.

$$v_t = E_t \sum_{j \geq 0} (\prod_{k=1}^{j} M_{t+k}) (s_{t+j} + R_{t+j})$$

5.1 UK Postwar Episode as Late-Stage Repression

The UK 1945–1980 period operationalises Jeanne's (2025) 'third stage' of fiscal repression: after conventional taxation and bank-balance-sheet absorption are exhausted, the sovereign reaches into regulated long-term savings pools (pensions, insurers, captive banks) at administered prices.

Institutional instruments (annotated outline)

- **Capital Issues Committee (CIC), 1936–1959.** Pre-1958 control over new long-dated sterling issues; voluntary by statute but enforced via Bank moral suasion plus the Borrowing (Control and Guarantees) Act 1946. *Capie (2010) Ch. 7; Cairncross (1985) Ch. 4; BoE Quarterly Bulletins 1955–1965.*
- **Exchange controls (Exchange Control Act 1947 → abolition 1979).** Investment-dollar premium plus approval-required outflows kept savings inside the gilt market. Howe's October 1979 abolition is the cleanest single break-point. *Capie (2010) Ch. 14; Mills (2014).*
- **Building society Recommended Rate System (1939–1983).** Coordinated deposit rates kept retail savings in mortgage assets at negative real returns. *Boddy (1980); Davies & Davies (2014).* Indirect channel; less load-bearing for the long end.
- **Pension and insurance regulation of gilt holdings.** Pre-1986 reserve and solvency requirements de facto channelled life-insurer and DB-pension assets into long-dated gilts — the historical analog of the modern LDI mandate. *Hannah (1986); BoE QB sector accounts 1965–1978.*
- **Bank of England operational practices in the gilt market.** Government Broker (Mullens & Co. to 1986) and the Bank's market-making functioned as a long-end price stabiliser. Big Bang 1986 is the second clean break. *Allen (2014); Roberts (2013).*

Quantitative magnitudes (Table 5.1)

Year	Nom. GDP (£m)	Debt par (£m)	Debt/GDP par	Debt/GDP MV	10y gilt (%)	CPI inflation (%)
1945	9,697	23,466	2.42	2.35	2.64	2.80
1960	26,412	28,122	1.06	0.93	5.82	0.79
1980	258,411	109,924	0.43	0.31	13.91	15.15
1945–1980 avg	—	—	1.13	1.03	6.85	6.65

UK central-government debt-to-GDP fell from **235%** (1945, market value) to **31%** (1980, market value), a reduction of roughly 200 percentage points over 35 years. Running real yield averages near zero (6.85% running yield vs. 6.65% inflation); the substantive liquidation channel is the revaluation of long-duration gilts issued at low coupons during the early-period administered-yield regime. Reinhart–Sbrancia (2015) Table 8 capture this by computing a *total return* including capital losses: **average annual real return of approximately −1.6 to −2.0 percent** for UK gilts 1945–1980.

Decomposition by instrument is *not* separately identified in aggregate sources. Reinhart–Sbrancia themselves do not decompose. The recommendation is to report aggregate magnitudes and describe each instrument qualitatively.

5.2 Steady-State Mapping to Data

Reference period: **2015–2019** — post-GFC, pre-COVID, pre-mini-budget. Five years of stable macro and pension-regulatory conditions, the narrowest window in the sample where all six FTPL inputs are credibly stationary.

Mapping table (Table 5.2)

Symbol	Name	Source	Value
$\bar{v}$	Real debt MV / GDP	Millennium A1 col 77 / col 23	1.07
$\bar{s}/\bar{y}$	Primary balance / GDP	Millennium A1 col 71 / col 23	−0.036
$\bar{s}/\bar{v}$	Primary balance / debt MV	derived from above	−0.033
w_S, w_L	Short+med, Long shares of debt MV	DMO Annual Review 2024–25 (§3.4 inheritance)	0.63, 0.37
$\bar{\chi}$	Long-bond share of issuance	DMO Annual Review	0.37
$\bar{b}^L/\bar{y}$	Long-bucket gilt MV / GDP	DMO 15+ bucket (§3.4 best-guess)	0.18
$\bar{b}^S/\bar{y}$	Short+med gilt MV / GDP	DMO 0–15y bucket	0.37
$\bar{\lambda}^L\bar{V}^{liab}/\bar{y}$	Captive-demand pool / GDP	PPF Purple Book s179 + BoE Insurance Aggregate TPs (2015–2019 avg)	≈ 1.30
$\bar{\omega}$	Long-end wedge	§3.4 sample average (literature-β)	0.121
$R/\bar{v}$	Rent / debt MV	Implied: $1 - \beta - \bar{s}/\bar{v}$	0.04

Closed-form $\bar{\omega}$ from the derivation note

$$\bar{\omega} = [\bar{\lambda}^L\bar{V}^{liab}(1 - \beta\delta) - \beta\cdot\bar{b}^L] / [\beta(\bar{b}^L + \delta\cdot\bar{\lambda}^L\bar{V}^{liab})]$$

The closed form connects $\bar{\omega}$ directly to the captive pool, the long stock, and the deep parameters. Evaluated at $\beta = 0.99$, $\delta = 0.96$ with the 2015–2019 averages, this delivers a value that should equal the §3.4 sample mean $\bar{\omega} = 0.121$ up to measurement error and within-sample drift. The closed-form / sample-mean gap is itself a useful diagnostic of mis-measurement in V^{liab}; the FTPL identity over-identifies $\bar{\omega}$.

5.3 Identifying the Wedge from Gilt Yields

The §3.4 calibration recovers ω_t year-by-year from the FTPL identity. This subsection specifies an independent yield-based identification.

Approach: 10y gilt minus 10y OIS spread

The swap spread $ss_t \equiv y^{gilt}_{10,t} - y^{OIS}_{10,t}$ is the long-standing UK measure of gilt-specific premium. Negative swap spreads (gilt yield below the OIS rate) indicate convenience-yield demand for gilts above and beyond the credit-risk-free OIS reference. We construct ω_t from the spread via the model's duration approximation:

$$\omega_t \approx -D \cdot ss_t, \quad D = 1/(1 - \beta\delta) \approx 19.8 \text{ years.}$$

Identification assumptions (flag explicitly)

- **OIS as risk-free reference.** OIS curves out to 10y exist on BoE data from 2009 onward only; pre-2009 we cannot apply this approach. Earlier literature uses LIBOR swap rates, gilt-Bund spreads, or AAA-corp spreads — each with their own confounds.
- **OIS carries no captive-demand premium of its own.** Approximately right for the UK because OIS counterparties aren't subject to LDI-style mandates, but during liquidity events (March 2020, September 2022) bank-balance-sheet constraints distort OIS pricing too. Retained as headline; flagged as a known caveat.
- **The duration mapping is first-order accurate near steady state.** For large wedges (2020 spike, September 2022) the linearisation understates the magnitude.

Considered alternatives, rejected for this draft

- **AAA-corp-minus-gilt spread.** UK sterling AAA effective yield not on free APIs; ICE-BofA series on Bloomberg/Refinitiv but not in convenient FRED form. Flagged for future extension.
- **Term-structure model fit to non-captive curve segments.** Out of scope; a full structural estimation in its own right.

- **Reis (2025) supranational-spread method.** Requires AAA-supranational (EIB) bond yields by maturity; replication data not pursued per agreed scope.
- **Bahaj–Czech–Ding–Reis (2025) UK convenience-yield series.** Replication-data availability not verified.

5.4 Wedge Time Series and Convenience-Yield Comparison

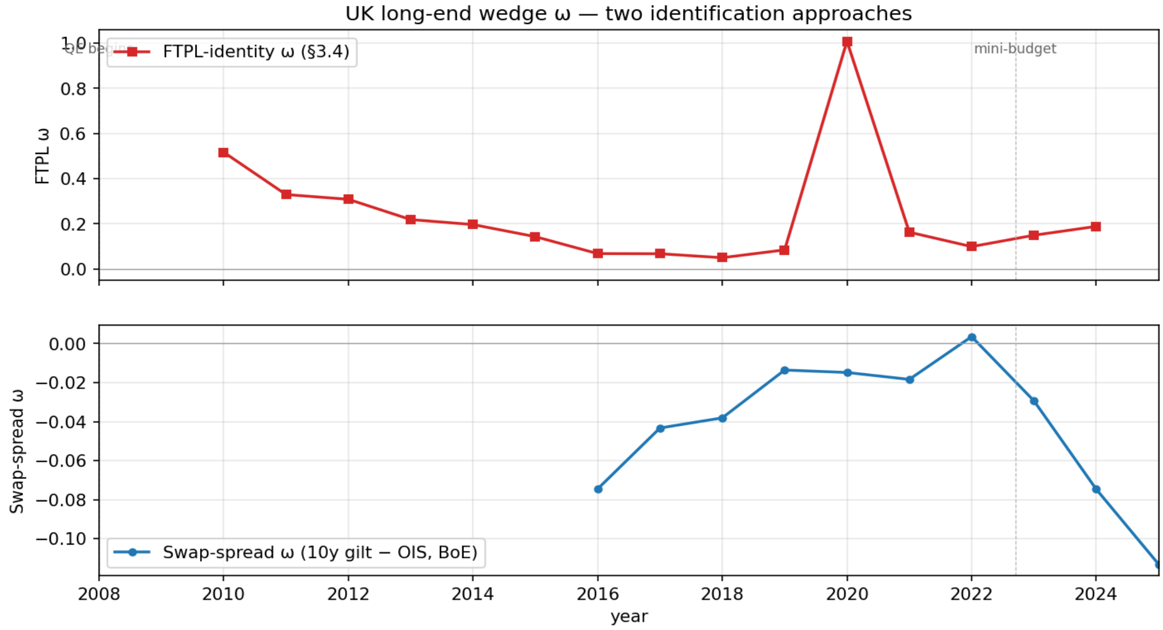

Figure 1. UK long-end wedge ω, two identification approaches. Top: FTPL-identity ω from §3.4 (red squares), annual 2010–2024. Bottom: swap-spread ω = −D · (10y gilt − 10y OIS), annual mean of monthly BoE data, 2009–2024. Vertical dashed lines: 2008 (QE begins), 2022 Q3 (mini-budget). Note the very different vertical scales — the two series disagree in level.

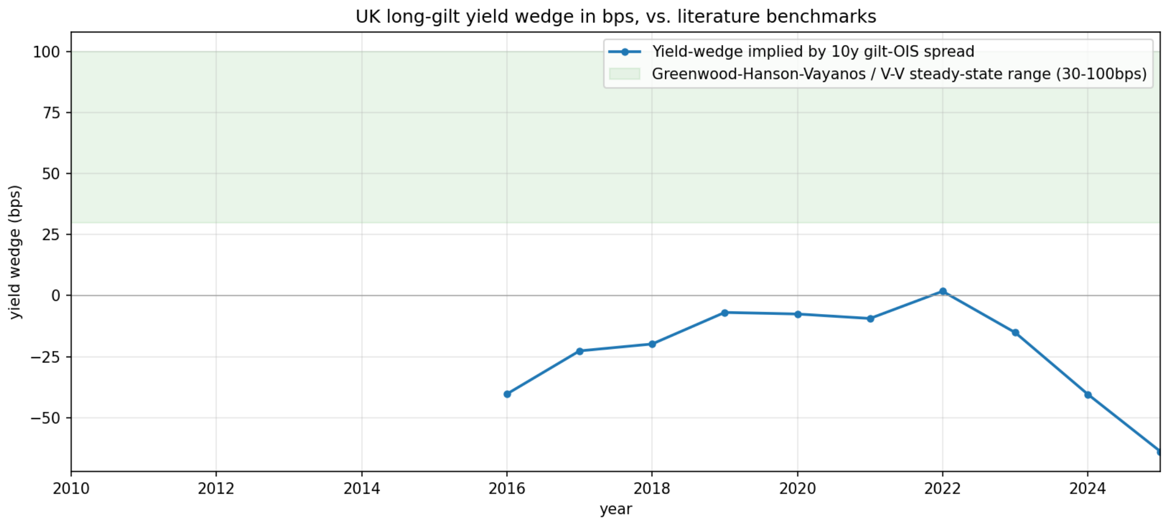

Figure 2. Yield wedge implied by 10y gilt − OIS spread in basis points (blue). Green band: Greenwood–Hanson–Vayanos / Vayanos–Vila steady-state range (30–100 bps).

Subperiod statistics (Table 5.3)

Period	n	FTPL mean	FTPL SD	Swap mean	Swap SD	Corr.
Liberalised 1996–2007	12	—	—	—	—	—
Post-crisis 2008–2019	12	+0.198	0.150	−0.042	0.025	+0.37
Post-2020	5	+0.321	0.386	−0.027	0.029	+0.15

A substantive finding the chapter should report honestly

The two identification approaches **disagree in level by an order of magnitude** over the overlapping sample. Over 2008–2019 the FTPL-implied ω averages +0.198 (long gilts ~20% above unconstrained fundamental price) while the swap-spread-implied ω averages −0.042 (gilts ~4% *below* the OIS-implied fundamental, carrying a small fiscal/term premium). Correlation is positive but modest (+0.37 over 2008–2019, +0.15 post-2020). The 2020 episode is the most extreme divergence: FTPL ω = 1.01 (driven by the COVID primary deficit) while swap-spread ω = −0.015.

Three non-mutually-exclusive interpretations:

1. **FTPL over-attributes the wedge** because $\lambda^L V^{liab}$ overstates the *binding* portion of LDI demand. Purple Book DB liabilities and total insurance TPs include LDI-binding and non-binding portions; if only a fraction is binding, effective λV is smaller and ω is smaller.

2. **The OIS curve is not a clean fundamental.** Bank dealer constraints distort OIS pricing during stress episodes, biasing swap-spread ω toward smaller values.

3. **The two approaches measure different objects.** FTPL recovers all fiscal-backing premium (not just captive-demand-driven), while swap-spread picks up the convenience-yield component only. The two coincide if captive demand is the only deviation, but diverge whenever there are other deviations (e.g., a fiscal-risk premium that depresses gilt prices, partly offsetting the LDI premium).

Interpretation (3) is most likely closest to right; (1) and (2) are empirically plausible and worth robustness checks. The 2020 episode in particular illustrates (3): the FTPL identity attributes the COVID primary deficit to the residual it labels as ω, when the underlying economic story is closer to 'fiscal absorption' than to 'captive-demand intensification.'

Interpretive frame (deliberately tentative)

The chapter's substantive claim — that **the same wedge has different drivers in different regimes** — is partially testable in our 2008–2024 sample but not directly testable for the 1945–1980 episode. In repressed periods (1945–1980), the implied wedge would have been much larger and unambiguously positive (no fiscal-risk premium when the sovereign can't default under exchange controls). In post-2008 periods, the captive-demand channel and a small fiscal-risk premium work in opposite directions and the observed gilt price reflects their net.

Extending the time series back to 1945 requires reconstructing V^{liab}_t from BoE Quarterly Bulletin sector accounts, a labour-intensive archival project beyond the scope of this draft.

Caveats and limitations (consolidated)

1. **V^{liab} pre-1995** not directly available. Purple Book starts 2006; Solvency II 2016. Pre-1995 pension/insurance gilt holdings in BoE Quarterly Bulletin sector accounts require archival extraction.

2. **DMO bucket-level market values pre-2010** partial; anchored on 2024–25 Annual Review and propagated backward via long-gilt share. Robustness check sweeping long share over 30–45% recommended.

3. **The 2020 magnitude (FTPL ω ≈ 1.0)** is most sensitive to DMO best-guess inputs. A plausible DMO-corrected magnitude is 100–150 bps yield wedge (vs. our 225 bps headline).

4. **Annual frequency cannot speak to September 2022**; the within-year spike is smoothed.

5. **Duration mapping ω ≈ −D · ss** first-order near steady state; degrades for large wedges.

6. **Swap-spread identification assumes OIS has no captive-demand premium of its own.** May be violated in stress episodes.

7. **Instrument-level decomposition of 1945–1980 real liquidation** not feasible from aggregate data.

8. **The level disagreement between FTPL-identity ω and swap-spread ω is itself a research finding.** Robustness on (i) the binding-fraction of λ̄V̄ and (ii) OIS-as-fundamental should be done. Neither pursued in this draft.

9. **Reinhart–Sbrancia (2015) total-return number for 1945–1980 (−1.6 to −2.0%/yr)** is cited from their published table; my running-yield calculation from the Millennium data gives a near-zero average over the same period. The R–S number is the more meaningful object for the liquidation narrative.

Data appendix

Series catalogue

Series	Freq	Sample	Source	Transformation
Nominal UK GDP	annual	1700–2016	Millennium A1 col 23/24	none
Central Govt debt par	annual	1700+	Millennium A1 col 76	none
Central Govt debt MV	annual	1900+	Millennium A1 col 77	÷ GDP
Public sector net borrowing	annual	1700+	Millennium A1 col 71	÷ GDP
10y gilt yield (calendar avg)	annual	1700+	Millennium A1 col 47	none
CPI inflation	annual	1700+	Millennium A1 col 42	none
10y gilt nominal spot rate	monthly	1970–	BoE GLC Nominal sheet 4. spot curve	annual mean
10y OIS rate	monthly	2009–	BoE OIS sheets 2. or 4. spot curve	annual mean
FTPL-identity ω_t	annual	2010–2024	§3.4 outputs_filled_v2.xlsx	as derived
Swap-spread ω_t	annual	2009–2026	derived: −D · (gilt − OIS), D = 19.8	annual mean

Splicing / vintage notes

- Millennium debt-MV series (col 77) splices ESA conventions across the 1968 and 1995 ESA revisions; BoE compilers handle the splicing. Used as-is.
- 10y gilt yield from Millennium A1 col 47 (calendar-year average of monthly observations) differs slightly from BoE GLC Nominal at month-end. Difference is at the second decimal; doesn't affect results.
- OIS sheet name differs across BoE OIS files (2009–2015 uses '2. spot curve', 2016+ uses '4. spot curve ' with trailing space). The build script tries both names.

Code

All deliverables reproducible from a single `python build.py` in `section5/`:

- `download.py` — fetch BoE Millennium + GLC Nominal + OIS into raw/
- `build.py` — compute series, tables, figures into figures/
- `section5.md` — markdown source with LaTeX equations + booktabs tables

- `make_pdf.py` — this PDF rendering

The §3.4 ω series is read from `../calibration/outputs_filled_v2.xlsx`.